InvenTrack

Inventory Management Platform

Product Requirements Document | v1.0

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1. Executive Summary

InvenTrack is a full-featured, cloud-native inventory management platform designed to compete directly with market leaders such as Zoho Inventory, Cin7, Fishbowl, and NetSuite. Based on comprehensive market research spanning the top-reviewed inventory tools of 2025–2026, InvenTrack consolidates the most critical capabilities demanded by small-to-mid-sized businesses (SMBs) and growing enterprises into a single, cohesive product.

> 📊 *Market research shows that 43% of companies still rely on spreadsheets for inventory — representing a massive addressable market for a modern, intuitive, and affordable cloud IMS.*

The platform targets the following primary verticals: retail (omnichannel), wholesale/distribution, light manufacturing, and eCommerce. InvenTrack will be offered as a SaaS product with tiered pricing, a free starter plan, and a RESTful API for enterprise customization.

2. Market Research Summary

2.1 Competitive Landscape

After researching the top-rated inventory management tools across Capterra, G2, GetApp, TechRepublic, and industry review sites as of Q1 2026, the following leaders were identified:

Product	Best For	Key Strength	Weakness	Pricing Model
Zoho Inventory	SMBs, eCommerce	Zoho ecosystem, automation	Limited scalability	Free + $29/mo+
Cin7	Omnichannel retail	Multi-channel sync	Complex setup	$349/mo+
NetSuite	Enterprise	ERP depth, forecasting	Very expensive	Custom
Fishbowl	Manufacturing	QuickBooks integration	Outdated UI	$4,395 one-time
Katana	Manufacturing SMBs	Production management	Narrow vertical	$179/mo+
Square for Retail	Small retail	Built-in POS + payments	Limited warehouse	Free + fees
GOIS	Growing businesses	AI forecasting	Newer, less proven	$Custom

2.2 Key User Pain Points (from Reviews)

Analysis of thousands of verified user reviews surfaced the following recurring pain points across existing tools:

- Steep learning curves for complex features (especially in NetSuite and Cin7)
- Poor or costly integrations with external eCommerce platforms and shipping carriers
- Multi-currency handling discrepancies requiring manual correction
- Limited or non-customizable reporting without purchasing add-ons
- Pricing that scales poorly — sudden cost jumps as businesses grow
- Mobile apps that lag behind the web interface in functionality
- Sync delays between sales channels and inventory (not truly real-time)

2.3 Feature Frequency Analysis

The following features appear across 90%+ of reviewed top-tier products and are considered table-stakes for InvenTrack:

- Real-time multi-location inventory tracking
- Automated purchase order generation based on reorder points
- Barcode / QR code scanning (mobile and desktop)
- Sales order and invoice management
- Shipping carrier integrations (multi-carrier label generation)
- Batch and serial number tracking
- Role-based access control (RBAC)
- Dashboard with KPI widgets and customizable reports
- eCommerce platform integrations (Shopify, Amazon, Etsy, WooCommerce)
- Accounting integrations (QuickBooks, Xero, Zoho Books)

3. Product Vision & Goals

3.1 Vision Statement

🎯 *InvenTrack gives every business — from a solo founder to a 500-person operation — the real-time inventory intelligence they need to sell confidently, restock proactively, and grow without friction.*

3.2 Strategic Goals

1. Launch an MVP that competes feature-for-feature with Zoho Inventory within 12 months.
2. Capture 1,000 paying customers in the SMB segment within 18 months post-launch.
3. Achieve 4.5+ star ratings on G2 and Capterra within the first year.
4. Provide a public API and webhooks to enable third-party integrations from day one.
5. Offer the most competitive free tier in the market (250 orders/month vs Zoho's 20).

4. Target Users & Personas

Persona	Role	Primary Need	Key Frustration
Sam the Shopkeeper	Retail store owner	See stock levels across 2 locations instantly	Spreadsheets & manual reconciliation
Elena the eCommerce Manager	DTC brand operator	Sync inventory across Shopify + Amazon	Overselling due to sync delays
Marco the Warehouse Lead	Warehouse supervisor	Pick-pack-ship with barcode scanning	Paper-based pick lists and errors
Priya the Finance Controller	Accountant / CFO	Real-time COGS, valuation & audit trail	Disconnected inventory and accounting data
Dev the System Integrator	IT / developer	RESTful API + webhooks for custom workflows	Closed systems with no extensibility

5. Product Scope

5.1 In Scope — MVP (Phase 1)

- Item & SKU management with variants, bundles, and composite items
- Multi-warehouse stock tracking with real-time updates
- Sales order lifecycle: create → approve → fulfill → invoice → payment
- Purchase order lifecycle: create → send to vendor → receive → bill
- Barcode scanning (web via camera, mobile app, USB scanner)
- Reorder point rules with auto-PO generation
- Batch and serial number tracking
- Role-based access control (Admin, Manager, Warehouse Staff, Accountant, Read-Only)
- Dashboard with 10 customizable KPI widgets
- 20+ built-in reports with CSV/Excel/PDF export
- Shopify and WooCommerce integrations
- QuickBooks Online and Xero integrations
- Multi-carrier shipping (FedEx, UPS, USPS, DHL) with label printing
- Mobile apps for iOS and Android
- RESTful API v1 with OAuth 2.0 authentication
- Webhooks for key events (stock alert, order created, PO received)

5.2 In Scope — Phase 2 (Post-MVP)

- Amazon, Etsy, eBay marketplace integrations

- Demand forecasting using historical sales trends
- Landed cost tracking (freight, duties, insurance)
- Vendor portal for self-service PO confirmation
- Advanced WMS: bin/shelf/zone management
- Cycle count scheduling and discrepancy workflows
- Multi-currency support with live FX rates
- Custom fields and custom modules

5.3 Out of Scope

- Full ERP modules (Payroll, HR, CRM) — integration-only approach
- Built-in POS hardware — will integrate with Square and Stripe Terminal
- Manufacturing MRP / Bill of Materials (Phase 3 consideration)

6. Detailed Feature Requirements

6.1 Item Management

The item catalog is the foundation of the platform. All stock, orders, and reports derive from items.

Feature	Description	Priority
Item profiles	Name, SKU, description, images (up to 10), weight, dimensions, cost price, selling price, tax rates, HSN/SAC code	P0
Product variants	Define attributes (color, size, material) generating variant matrix with per-variant SKUs and pricing	P0
Composite items / Bundles	Group items into sellable bundles; auto-deduct component stock on sale	P0
Units of measure	Define primary and secondary UoM with conversion rates (e.g., 1 case = 24 units)	P1
Item groups	Organize items into hierarchical categories and sub-categories	P0
Low-stock thresholds	Per-item, per-warehouse reorder point and reorder quantity settings	P0

Feature	Description	Priority
Item activity log	Full audit trail of stock movements, edits, and transfers per item	P0
Bulk import/export	CSV and Excel import with field mapping wizard; full export in same formats	P0

6.2 Inventory Tracking

InvenTrack must maintain accurate, real-time inventory counts across all locations with full traceability.

Feature	Description	Priority
Multi-warehouse tracking	Separate stock counts per warehouse location; global availability view	P0
Stock adjustments	Manual +/- adjustments with reason codes and mandatory notes field	P0
Inter-warehouse transfers	Transfer orders with source/destination, quantity, and shipment tracking	P0
Batch tracking	Assign batch/lot numbers; track by expiry date; FIFO/FEFO fulfillment	P0
Serial number tracking	Unique serial per unit; full history from receipt to sale	P0
Stock valuation	Support FIFO, LIFO, and Weighted Average costing methods	P1
Cycle counting	Schedule partial counts by category or location; variance reporting	P1
Barcode scanning	Scan to receive, adjust, transfer, and fulfill using device camera or USB scanner	P0

6.3 Order Management

Sales and purchase order workflows must be configurable, multi-step, and fully auditable.

6.3.1 Sales Orders

- Create sales orders manually or auto-generate from integrated sales channels
- Order statuses: Draft → Confirmed → Picked → Packed → Shipped → Invoiced → Paid → Closed
- Pick list generation optimized by warehouse bin location
- Packing slip and delivery note PDF generation
- Backorder handling: fulfill partial quantities, auto-create backorder for remainder
- Dropshipping: route order directly to vendor for fulfillment
- Customer portal for order status tracking (Phase 2)

6.3.2 Purchase Orders

- Create POs from reorder alerts, manual entry, or vendor catalog
- PO statuses: Draft → Sent → Partially Received → Fully Received → Billed → Closed
- Goods receipt with quantity and quality check workflows
- PO approval routing: configurable approval chains by PO value thresholds
- Vendor price lists and payment terms management
- Email POs directly to vendors from the platform with PDF attachment

6.4 Shipping & Fulfillment

Feature	Description	Priority
Multi-carrier rates	Compare real-time rates from FedEx, UPS, USPS, DHL in checkout	P0
Label printing	Generate and print shipping labels in ZPL (thermal) and PDF (laser) formats	P0
Tracking sync	Auto-sync tracking numbers to sales orders; notify customers by email/SMS	P0
AfterShip integration	Universal tracking via AfterShip for 900+ carriers	P1
Batch label printing	Print labels for multiple orders in a single batch job	P1
Shipment manifests	Generate end-of-day manifests per carrier	P1

6.5 Reporting & Analytics

Reports must be fast, filterable, exportable, and schedulable. Key reports include:

- Inventory Summary — current stock on hand, on order, committed, available
- Stock Movement Report — all ins and outs for any date range
- Inventory Valuation Report — total stock value by FIFO/LIFO/WAC method
- Sales by Item Report — units sold, revenue, COGS, gross margin per SKU
- Purchase Order History — POs by vendor, date range, status
- Reorder Report — items below reorder threshold with recommended PO quantities
- Batch/Serial Traceability Report — trace any unit from receipt to sale
- Warehouse Performance Report — transfer efficiency, receiving speed
- Demand Forecast Report (Phase 2) — projected stock needs by SKU
- Dead Stock Report — items with zero movement for configurable days threshold

📋 *All reports must support: date range filters, warehouse filters, item/category filters, column customization, save-as-custom-view, and export to CSV, Excel (.xlsx), and PDF.*

6.6 Dashboard

The main dashboard provides at-a-glance visibility into business health. All widgets are drag-and-drop configurable.

Widget	Data Displayed
Stock Alerts	Items below reorder point, grouped by urgency (critical / low)
Sales Overview	Orders this week vs. last week, revenue trend sparkline
Purchase Orders	Open POs count, total $ value, expected delivery timeline
Top Selling Items	Top 10 SKUs by units sold in selected period
Inventory Value	Total on-hand value with trend vs. prior period
Pending Shipments	Orders ready to ship, overdue, and in transit counts
Vendor Performance	Average lead time and on-time delivery rate per vendor
Recent Activity	Chronological feed of all stock movements and order events

6.7 Integrations

InvenTrack uses a webhook-based, bi-directional sync architecture. All integrations sync on a maximum 5-minute polling cycle, with push webhooks for real-time critical events.

Category	Integrations — Phase 1	Integrations — Phase 2
eCommerce	Shopify, WooCommerce	Amazon Seller Central, Etsy, eBay, BigCommerce
Accounting	QuickBooks Online, Xero	Zoho Books, FreshBooks, Sage
Shipping	FedEx, UPS, USPS, DHL, AfterShip	Shiprocket, EasyPost, ShipStation
Payments	Stripe, PayPal	Razorpay, Square
Communication	Email (SMTP), SMS (Twilio)	Slack, WhatsApp Business
Custom	REST API + Webhooks	Zapier / Make connector

7. Non-Functional Requirements

Category	Requirement	Target
Performance	API response time (95th percentile)	< 300ms
Performance	Dashboard load time (cold)	< 2 seconds
Performance	Bulk import (10,000 items)	< 60 seconds
Availability	Platform uptime SLA	99.9% (8.7 hrs/year max downtime)
Scalability	Concurrent users per tenant	500+ without degradation
Security	Authentication	OAuth 2.0, MFA, SSO (SAML 2.0 — Phase 2)
Security	Data encryption	AES-256 at rest, TLS 1.3 in transit
Security	Audit trail	Immutable log of all data changes with user, timestamp, diff
Compliance	Data residency options	US, EU, India (AWS multi-region)
Mobile	iOS and Android app parity	Core features at launch; full parity within 6 months
Accessibility	Web app WCAG compliance	WCAG 2.1 AA

8. Recommended Tech Stack

Layer	Technology	Rationale
Frontend	React 19 + TypeScript + Tailwind CSS	Component reuse, type safety, rapid UI development
State Management	Zustand + React Query (TanStack)	Lightweight global state + server state caching
Backend API	Node.js (Fastify) or Python (FastAPI)	High throughput; rich ecosystem for integrations
Database	PostgreSQL (primary) + Redis (cache)	ACID transactions for inventory; sub-ms caching
Search	Elasticsearch or MeiliSearch	Fast full-text item/order search across large catalogs
Message Queue	BullMQ (Redis-backed)	Async job processing (label gen, reports, syncs)
File Storage	AWS S3 + CloudFront CDN	Item images, export files, generated PDFs
Auth	Auth0 or Clerk	OAuth 2.0, MFA, social login out-of-the-box
Infrastructure	AWS ECS (Fargate) + RDS + ElastiCache	Serverless containers; managed DB and cache
CI/CD	GitHub Actions + Docker	Automated test, build, deploy pipeline
Monitoring	Datadog or Grafana + Sentry	APM, error tracking, uptime alerts
Mobile	React Native (Expo)	Code sharing with web React; fast iteration

9. Pricing Model

InvenTrack's pricing is designed to undercut Zoho Inventory and Cin7 at every tier while offering higher limits.

Plan	Price	Orders/Month	Warehouses	Users	Key Features
Free	$0	250	1	2	Core inventory, 1 integration, community support
Starter	$29/mo	1,500	2	5	All Free + shipping, barcode, API access

Plan	Price	Orders/Month	Warehouses	Users	Key Features
Growth	$79/mo	10,000	5	15	All Starter + batch/serial, advanced reports, webhooks
Pro	$199/mo	Unlimited	Unlimited	Unlimited	All Growth + forecasting, custom fields, priority support
Enterprise	Custom	Unlimited	Unlimited	Unlimited	All Pro + SSO, SLA, dedicated CSM, custom integrations

10. Development Roadmap & Milestones

Milestone	Timeline	Deliverable
M0 — Foundation	Weeks 1–4	Repo setup, DB schema, auth, CI/CD, design system
M1 — Inventory Core	Weeks 5–10	Item catalog, multi-warehouse tracking, barcode scanning
M2 — Order Management	Weeks 11–18	Sales orders, purchase orders, fulfillment workflows
M3 — Integrations	Weeks 19–24	Shopify, QuickBooks, shipping carriers, email/SMS
M4 — Reporting	Weeks 25–28	20+ reports, dashboard widgets, CSV/PDF export
M5 — Mobile Apps	Weeks 25–30	iOS and Android apps (barcode, orders, adjustments)
M6 — Beta Launch	Week 32	Closed beta with 50 customers, feedback collection
M7 — GA Launch	Week 40	Public launch, all Phase 1 features, billing active
M8 — Phase 2	Weeks 41–60	Forecasting, Amazon, multi-currency, WMS bins

11. Success Metrics

11.1 Launch KPIs (Month 6 post-GA)

- 1,000 registered accounts (free + paid)
- 150 paying customers
- < 5% monthly churn on paid plans
- NPS score > 45
- Average API response time < 200ms (p95)

11.2 Year 1 KPIs

- $500K ARR
- 4.5+ stars on G2 and Capterra (minimum 100 reviews each)
- 3 enterprise contracts (> $20K ACV each)
- 99.95% uptime achieved
- 50+ third-party integrations via public API

12. Open Questions & Decisions Required

6. Should the mobile app be built in React Native (Expo) or as a separate native app per platform? Recommendation: React Native for speed-to-market.
7. Which costing method should be the default? Recommendation: Weighted Average, with option to switch per-account in settings.
8. Should barcode printing (label design) be in-app or via integration with Dymo/Zebra software? Recommendation: In-app template builder (Phase 2); export ZPL for Phase 1.
9. What is the data retention policy for audit logs? Recommendation: 7 years for compliance with most jurisdictions.
10. Should we build a public marketplace for third-party InvenTrack plugins (like Shopify's App Store)? Recommendation: Evaluate at Month 12 post-launch.

Appendix A: Glossary

SKU	Stock Keeping Unit — a unique identifier for a specific product variant
PO	Purchase Order — an order sent to a vendor to replenish stock
WMS	Warehouse Management System — software that manages physical warehouse operations
FIFO	First In, First Out — costing method where oldest stock is sold first
FEFO	First Expired, First Out — fulfillment rule prioritizing soonest-to-expire batches
COGS	Cost of Goods Sold — direct costs of producing sold inventory
Reorder Point	The stock level that triggers a new purchase order

Backorder	An order for a product that is temporarily out of stock
Pick List	A warehouse document listing items and locations to collect for order fulfillment
ARR	Annual Recurring Revenue — total annualized value of subscription contracts

End of Document — InvenTrack PRD v1.0